
THE FUNDING CARRY AND A CROSS-VENUE SPREAD ON PERPETUAL FUTURES: A SIGNIFICANCE-TESTED STUDY OF HYPERLIQUID AND CENTRALIZED VENUES

Tony Lau
York University

ABSTRACT

Hyperliquid has rapidly become the dominant decentralized venue for perpetual futures (over US\$6 billion in total value locked as of June 17, 2026), making the economics of its distinctive hourly funding mechanism increasingly consequential. Our central finding is a robust, economically sensible edge: Hyperliquid’s funding structurally exceeds that of major centralized venues (Binance and Bybit, both over the full three-year sample) on the same assets by roughly 7% annualized (14.7% in 2024, 4.6% in 2025)—a delta-neutral cross-venue dislocation (long the centralized-venue perpetual, short the Hyperliquid perpetual) that is highly significant (Newey–West HAC $t = 4-8$), walk-forward consistent (4–5 of 5 folds), survives the cross-venue basis, capital, and transaction costs when held statically—returning roughly 7–11% on deployed capital at a prudent $2-3\times$ per-leg leverage, with negligible liquidation risk—and, unlike the single-venue premium, remains statistically significant into 2026.

The single-venue carry—long spot, short the perpetual—is a general phenomenon, present on every venue but far richer on Hyperliquid: a genuine, highly *time-varying* risk premium of $\sim 7\%$ over the risk-free rate on Hyperliquid (17.9% in 2024, 3.6% in 2025) versus only $\sim 2-3\%$ on Binance and Bybit—on notional, or $\sim 5\%$ on the capital actually deployed at a prudent $2-3\times$ short-leg leverage (Hyperliquid perps are USDC-margined). The methodology is validated by HAC paired t -tests, a block bootstrap, five-fold walk-forward, and a Deflated Sharpe that corrects for multiple testing; returns are measured in *excess of the risk-free rate* (the 3-month T-bill, which the on-chain Aave USDC supply rate tracks closely); and a factor-model alpha.

We further test a battery of “smart” refinements—cost-recovery exits, cross-sectional allocation, ML return-tilts and tail-risk leverage, and options tail-hedging—and all *fail* to beat a naive baseline, several significantly. We report these nulls as a feature: a cautionary catalog of holdout-selection artifacts that only out-of-sample, multiple-testing-corrected inference exposes.

1 INTRODUCTION

Hyperliquid has, in under three years, grown from a new decentralized exchange into the dominant on-chain venue for perpetual futures, with total value locked exceeding US\$6 billion as of June 17, 2026, and rapidly rising volume and open interest.¹ As this venue’s share of perpetual activity grows, whether its distinctive hourly funding mechanism creates exploitable, *durable* edges becomes materially important to researchers and practitioners alike.

The largest *centralized* venues—Binance and Bybit—remain far bigger by perpetual volume and open interest and effectively set the reference funding level for crypto perpetuals; they are the natural

¹Total value locked from DefiLlama, <https://defillama.com/protocol/hyperliquid>, as of June 17, 2026; by which date Hyperliquid intermediated the large majority of decentralized perpetual-futures volume.

comparison for any claim that Hyperliquid’s funding is rich. The funding *formulas* are in fact nearly identical: each venue sets the rate to the perpetual’s premium over the index plus a clamped interest term, $F = \text{premium} + \text{clamp}(r - \text{premium}, \pm 0.05\%)$, with the same interest anchor $r = 0.01\%$ per 8h ($\approx 11\%$ APR).² The formula tethers the expiry-less perpetual to spot through the premium term and charges the cost of carry through the interest term, with the clamp bounding the rate for stability. Two design choices differ: Hyperliquid charges *hourly* and caps funding loosely at 4%/hour, whereas Binance and Bybit charge every eight hours under tighter caps. Because the rules and interest anchor are shared, the gap we document is *not* an interest-rate artifact—nor (since both sum to a daily total before annualizing) a frequency artifact. It reflects Hyperliquid’s perpetual carrying a persistently higher premium over its oracle than the centralized venues do—which, the interest anchor being shared, is precisely what the cross-venue spread measures—consistent with stronger net long-side demand on the DEX, passed through by its loose cap. The underlying asset is identical; the dislocation is in the realized premium. (We measure this relative premium difference directly; we do not claim a market-wide directional bias, and funding is positive partly by construction through the interest term.)

We study these economics through the delta-neutral funding carry (long spot, short perpetual) and a cross-venue extension, and our headline result is positive and robust: Hyperliquid’s funding structurally exceeds that of major centralized venues (Binance and Bybit) on the same assets, a delta-neutral spread of roughly 7% annualized that is statistically significant on both venues over the full sample, walk-forward consistent, and—unlike the single-venue premium—still significant in 2026. At a prudent 2–3 $\times$ per-leg leverage the cross-venue trade returns roughly 7–11% on deployed capital (about 17% at 5 $\times$), with negligible liquidation risk.

The single-venue carry—long spot, short the perpetual—is itself a general phenomenon rather than a Hyperliquid peculiarity: it is present on the centralized venues too, but far richer on Hyperliquid. It is a real but highly time-varying risk premium of about 7% over the risk-free rate on Hyperliquid, versus only $\sim 2\text{--}3\%$ on Binance and Bybit. As a return on capital this $\sim 7\%$ is a *ceiling*, not a floor: higher short-leg leverage frees margin and lifts return-on-capital *toward* it, but the fully-paid spot leg—which always ties up roughly the notional—caps it at the funding yield. So, unlike the cross-venue spread (whose two margined-perp legs let leverage scale return-on-capital to $\sim 17\%$ at 5 $\times$), the carry cannot exceed $\sim 7\%$ without also leveraging the spot, and liquidation risk on the short binds the usable leverage at 2–3 $\times$, leaving a realized $\sim 5\%$: Hyperliquid perps are USDC-margined, so the spot leg ties up separate capital, and reaching the $\sim 7\%$ ceiling would require the full spot-perp cross-collateralization the venue is still building toward.

Equally central is our methodological stance. Crypto strategy research is rife with backtests that look compelling and evaporate out of sample, so we hold every claim to the standards normal in quantitative finance: pre-registration, Newey–West HAC tests, block bootstrap, walk-forward validation, a multiple-testing-corrected Deflated Sharpe, returns measured in excess of the risk-free rate, and a factor-model alpha. Under this lens a sequence of increasingly sophisticated overlays—cost-recovery exits, cross-sectional allocation, ML return-tilts and tail-risk leverage, options tail-hedging—almost all fail to beat a naive baseline, several significantly. We report these nulls as a feature: the recurring failure mode is an overlay tuned on a recent holdout that a walk-forward or bootstrap test reverses—a pattern only out-of-sample, multiple-testing-corrected inference reliably exposes.

2 DATA AND METHODOLOGY

We use hourly open/high price series, Hyperliquid funding rates and premia, and Binance funding rates and premium indices for eight assets (BTC, ETH, SOL, BNB, XRP, LINK, AVAX, DOGE), spanning mid-2023 to mid-2026 — roughly 26,600 hourly observations per asset ($\approx 213,000$ pooled;

²Worked example with Binance’s 8-hour rate. *Neutral* (premium 0): $F = 0 + \text{clamp}(0.01\%, \pm 0.05\%) = +0.01\%$, i.e. $\approx 11\%$ APR, longs paying shorts. *Rich* (premium +0.10%): $F = 0.10\% + \text{clamp}(-0.09\%, \pm 0.05\%) = +0.05\%$ ($\approx 55\%$ APR). *Sustained discount* (premium -0.10%): $F = -0.10\% + \text{clamp}(0.11\%, \pm 0.05\%) = -0.05\%$. So the interest term anchors funding near +11% APR across the near-spot regime, and funding turns negative only at a sustained perpetual discount. Hyperliquid uses the same formula and 0.01%/8h interest anchor, charged hourly (one-eighth each hour) and capped at 4%/hour, so the worked values are identical—which is why the realized cross-venue gap is in the premium, not the rule.

Table 1: Cross-venue funding spread, decomposed into its two legs: *HL recv.* is the funding collected on the short Hyperliquid perpetual, *Binance paid* the funding paid on the long Binance perpetual, and *Net spread* their difference (annualized). *NET Sharpe/maxDD* then book the cross-venue basis as daily mark-to-market (a tight $\sim 2.3\text{bp/day}$, so the realistic NET Sharpe barely falls below the funding-only ~ 9). All net spreads are highly significant (HAC $t = 4\text{--}8$, bootstrap $p = 0.00$) and walk-forward positive in 4–5/5 folds. (Leg columns are full-sample means and the net is the daily-aligned spread, so the displayed difference may differ by $\leq 0.1\text{pp.}$)

Coin	HL recv.	Binance paid	Net spread	Basis/day	NET Sharpe	NET maxDD
ETH	14.6%	7.6%	+7.1%	2.3bp	9.00	−1.15%
BTC	14.5%	7.3%	+7.2%	2.3bp	8.87	−1.89%
SOL	12.6%	5.1%	+7.5%	2.5bp	6.47	−3.63%
DOGE	17.4%	8.1%	+9.3%	2.6bp	5.49	−11.28%

$\approx 1,100$ trading days). The Deribit DVOL implied-volatility index is collected hourly for BTC and ETH. The carry return per asset is computed in a backtester that charges perpetual and spot fees and slippage, books funding and basis mark-to-market, and shifts signals one bar to avoid look-ahead.

This hourly resolution is what gives the level and spread tests their power. But it must be discounted correctly: intraday funding returns are strongly autocorrelated, so the effective sample is far smaller than the raw observation count. We therefore report Newey–West HAC standard errors and daily-resampled Sharpe ratios throughout, so that significance is not overstated by the autocorrelated hourly series. Consequently the binding limitation of this study is *not* the number of observations but the calendar span: the data cover approximately one post-FTX market regime (~ 3 years). This is why the level and spread significance are strong (HAC $t = 4\text{--}8$) while claims about the premium’s forward stability and any secular compression trend remain inconclusive — additional independent regimes, not additional hours, are what those questions require.

For significance we report, for each daily return stream, the annualized Sharpe ratio with a Lo (Lo, 2002) confidence interval, Sortino, Calmar, maximum drawdown, and the Probabilistic Sharpe Ratio. For strategy-versus-baseline comparisons we use a Newey–West (Newey & West, 1987) HAC paired t -test on the daily return difference, a block bootstrap for the Sharpe and APR difference, and a Deflated Sharpe Ratio (Bailey & López de Prado, 2014) to penalize the multiple configurations tried, in the spirit of the multiple-testing critique of the cross-sectional asset-pricing literature (Harvey et al., 2016). Every comparative claim is additionally subjected to a five-fold walk-forward consistency check, which we find to be the single most informative guard against overfitting.

Implementation. Returns are computed daily (hourly funding resampled to daily) and annualized with 365 periods, since crypto trades continuously. Newey–West HAC standard errors use the automatic lag $\lfloor 4(n/100)^{2/9} \rfloor$; the block bootstrap uses 10-day blocks with 3,000 resamples (95% confidence intervals and one-sided p -values). The Probabilistic and Deflated Sharpe ratios follow Bailey & López de Prado (2014), the latter deflating for the number of configurations tried within each overlay family (e.g. five for the cost-recovery-exit sweep); the Probabilistic Sharpe is benchmarked at a zero Sharpe with a skewness/kurtosis correction. VaR and CVaR are reported at the 95% level.

3 THE CROSS-VENUE FUNDING SPREAD: A ROBUST EDGE

Our central, most robust result is a cross-venue *dislocation*. Hyperliquid’s hourly funding structurally exceeds Binance’s eight-hourly funding by roughly 7% annualized, on 70–80% of days—despite the two venues sharing essentially the same funding formula and interest anchor (Introduction), so the gap is in the realized premium, not the rule. Shorting the Hyperliquid perpetual and going long the Binance perpetual on the same coin is delta-neutral and collects this spread.

This is consistent with reports that cross-venue funding markets are two-tiered and that CEX/DEX funding-rate arbitrage carries real but bounded profit (Zhivkov, 2026; Werapun et al., 2025). Table 1 shows the edge survives a realistic accounting. Because both perpetuals track the same coin, the cross-venue *basis* (Hyperliquid minus Binance premium) is tight ($\sim 2.3\text{bp/day}$), so adding it as daily

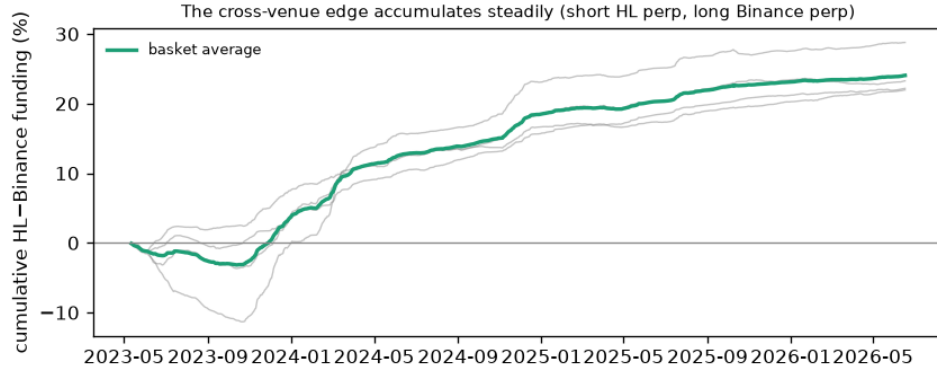


Figure 1: **The durable edge.** Cumulative Hyperliquid-minus-Binance funding (short HL perp, long Binance perp); grey lines are the four assets, green the basket average. After a brief early period the spread accumulates steadily to $\approx 24\%$ over the sample and persists into 2026—unlike the single-venue premium, whose 2026-to-date excess over the risk-free rate has compressed to near zero.

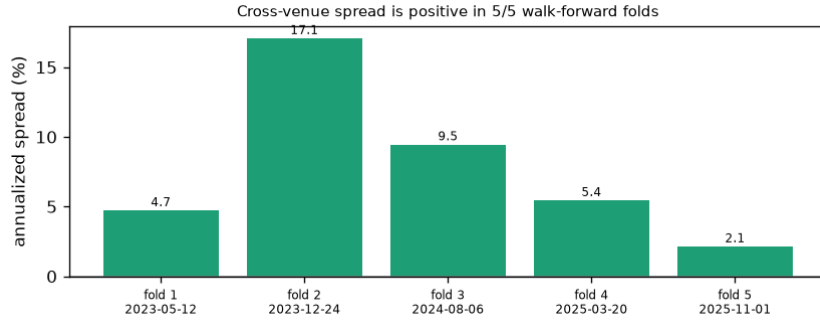


Figure 2: **Out-of-sample robustness.** The basket cross-venue spread is positive in all five chronological walk-forward folds (per-asset dynamic-Sharpe wins 4–5/5, Table 1). The fold heights also show the time-variation: the edge peaks in the 2024 fold (17%) and decays to $\sim 2\%$ by the 2025–26 fold, while staying positive throughout.

mark-to-market barely dents the Sharpe. The trade is best held *statically*—the spread is persistently positive, so one simply holds and collects, at negligible cost—whereas a dynamic flip-on-sign variant dies on transaction costs (negative APR beyond 20bp round-trip). It is capital-efficient: with margin required on both venues, return-on-capital is about 7%, 10%, and 17% at two-, three-, and five-times per-leg leverage, with leg-liquidations rare until roughly eight times. The majors are robust (drawdowns of 1–2%); DOGE is materially riskier. The spread is likewise time-varying (14.7% in 2024, $\sim 2\%$ in 2026), so the forward edge is well below the full-sample figure—but, notably, it is *more persistent than the single-venue carry premium*: it remains statistically significant in 2026 (2.0%, $t = 4.7$) whereas the carry premium over the risk-free rate has compressed to near zero in 2026 to date. As with the premium, a formal compression trend is not statistically significant over three years.

Why the gap is not arbitrated away. Funding is paid peer-to-peer (longs to shorts), not by the venue, so the persistent premium is what the marginal short demands to take the other side of the net long demand on Hyperliquid. Three frictions keep it open. First, *capital fragmentation*: the trade needs margin on both venues with no cross-venue netting, so it locks up roughly twice the notional and earns a return on capital well below the gross spread. Second, *venue risk*: Hyperliquid is a young on-chain protocol, and a delta-neutral position still bears smart-contract, oracle, liquidation-engine, and bridge/withdrawal risk—if Hyperliquid halts or fails, the hedge breaks. Third, the spread is *time-varying* and has compressed, so the forward edge is uncertain. The $\sim 7\%$ is therefore best read

as compensation for bearing Hyperliquid-specific risk and locking up fragmented capital—a limits-to-arbitrage premium that draws short-side liquidity to a newer venue—rather than a free lunch.

Robustness across venues. The premium is not specific to Binance. Table 2 compares Hyperliquid’s funding to four major centralized venues. The two venues that publish full history—Binance and Bybit—*both* show a structural Hyperliquid premium of roughly 6–8% annualized over the three-year sample, highly significant on every asset (HAC $t = 4-8$); measured against the cross-venue average the premium is +6.7 to +8.7% ($t = 4-8$), and against the *highest*-funding venue each day it is still +4.3 to +6.5% ($t = 3-6$). OKX and Bitget publish only the most recent ~ 3 months of funding history; over that window—which falls in the compressed 2026 regime documented above—Hyperliquid still exceeds them on a majority of days, but by the smaller margin consistent with the time-variation, not a weaker edge. That two independent full-history venues agree is what justifies calling the dislocation *structural*.

Table 2: Hyperliquid funding minus each major centralized venue’s funding (delta-neutral daily spread), averaged across BTC/ETH/SOL/DOGE; the HAC t column is the range across the four assets. The *Funding APR* column is each venue’s own gross funding (for reference Hyperliquid’s averaged 14.8% over the full sample); the spread is the funding *differential*, HL minus the venue’s, collected delta-neutral. Binance and Bybit publish full history; OKX and Bitget publish only ~ 3 recent (compressed-regime) months, so their figures are not comparable to the three-year columns. Per-asset detail is in the replication results.

Venue	Sample	Funding APR	HL—venue spread	% days HL higher	HAC t
Binance	3y (1,133d)	7.0%	+7.8%	74%	4.2–8.2
Bybit	3y (1,133d)	7.8%	+6.9%	70%	4.0–8.2
OKX	~ 3 mo (87d)	1.8%	+1.2%	59%	–0.5–2.3
Bitget	~ 3 mo (83d)	1.6%	+1.5%	62%	–1.2–3.1

Is the carry venue-specific? The single-venue carry exists on every venue, but is far richer on Hyperliquid. Table 3 runs the identical long-spot/short-perp construction on each venue with full data: the Hyperliquid carry earns a $\sim 10\%$ premium over the risk-free rate versus only $\sim 2-3\%$ on Binance and Bybit. The carry is thus *not* a generic perpetual-futures phenomenon—it is specific to Hyperliquid’s funding mechanism—and the gap is precisely the cross-venue spread: HL minus CEX carry premium ($10.1 - 3.1 \approx 7\%$) reproduces Table 2. (These are funding-only figures on the four cross-venue assets, so the Hyperliquid premium reads $\sim 10\%$ here; the paper’s headline $\sim 7\%$ net premium is the eight-asset basket after the full execution-cost model. The *ordering* and the decomposition, not the level, are the point.)

Table 3: Single-venue carry (long spot, short the named venue’s perpetual), funding-only, buy-and-hold, daily, basket of BTC/ETH/SOL/DOGE, in excess of the 3-month T-bill, which averaged 4.7% over the sample (so gross APR less $\sim 4.7\%$ gives the premium over rf). The Hyperliquid carry is roughly $4\times$ richer than the CEX carry, and the difference equals the cross-venue spread of Table 2. OKX/Bitget lack the history for a full carry.

Venue	Gross APR	Premium over rf	maxDD	% days positive
Hyperliquid	14.8%	+10.1%	–3.5%	81%
Bybit	7.8%	+3.1%	–0.7%	82%
Binance	7.0%	+2.3%	–1.0%	82%

A second tradeable construction. Because Bybit also publishes full funding and mark/index data, the realistic basis-and-cost test of Table 1 can be re-run end to end with a Bybit leg (short Hyperliquid, long Bybit). Table 4 shows near-identical economics: a tight cross-venue basis (3–4.5 bp/day), NET Sharpe of 7.6–8.2 on the majors, return-on-capital of about 7%, 10%, and 17% at two-, three-, and five-times per-leg leverage, and the same static-cheap/dynamic-dies-on-cost profile (static APR $\approx 6.8-8.1\%$ even at 40 bp; dynamic negative beyond 20 bp). The edge is thus *tradeable on two independent venues*, not an artifact of one exchange.

Table 4: Full tradeable construction on a second venue: short Hyperliquid perp, long Bybit perp, with the cross-venue basis booked as daily mark-to-market (Bybit premium = mark/index - 1). Directly comparable to Table 1.

Coin	Spread APR	Basis std/day	NET APR	NET Sharpe	NET maxDD
ETH	7.0%	3.0bp	7.0%	8.16	-1.27%
BTC	6.8%	3.1bp	6.7%	7.59	-2.00%
SOL	6.0%	4.3bp	6.0%	4.39	-4.73%
DOGE	8.1%	4.5bp	8.1%	4.74	-11.17%

4 THE FUNDING CARRY: A REAL BUT TIME-VARYING RISK PREMIUM

The cross-venue spread is, at bottom, the gap between Hyperliquid’s funding carry and the centralized venues’ (Table 3); we now study that single-venue carry on its own. Held delta-neutral, the carry earns a statistically significant return: in-sample APR near 14%, out-of-sample APR of roughly 6–7% after fees, with a Sharpe confidence interval excluding zero and a Probabilistic Sharpe Ratio near one. This is consistent with the crypto-carry literature (Schmeling et al., 2023), studies of Bitcoin-futures basis and arbitrage (De Blasis & Webb, 2022), and perpetual pricing theory (Ackerer et al., 2026; Kim & Park, 2025): funding is compensation for bearing risk—short-leg liquidation on up-moves, funding-regime reversals, and execution/basis frictions—not a free lunch. The headline Sharpe (of order 12 on idealized assumptions) is an upper bound; the honest figure is the 6–7% net APR on notional—about ~5% on the capital actually deployed at a prudent two-to-three-times leverage, since Hyperliquid perps are USDC-margined.

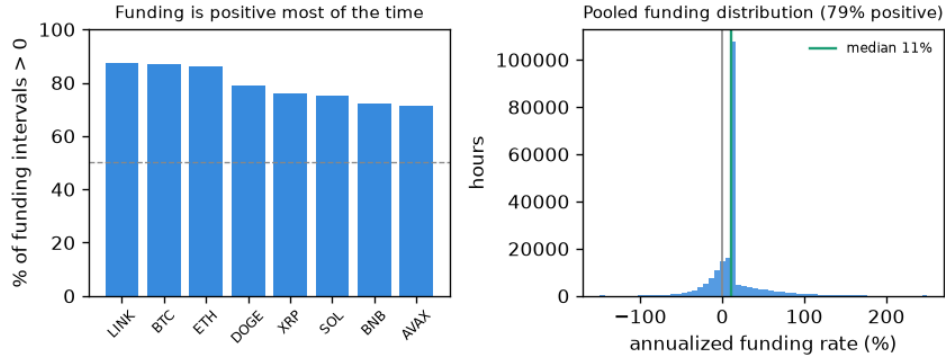


Figure 3: **The economic engine.** Left: the share of funding intervals with positive funding, by asset—on Hyperliquid, perpetual longs pay shorts 71–87% of the time. Right: the pooled distribution of annualized funding is strongly right-skewed (median $\approx 11\%$, 79% of hours positive). A delta-neutral short-perp/long-spot position harvests this persistent positive funding.

Crucially, this premium is an *average over a highly time-varying series*, not a stable level. Measured in excess of the 3-month T-bill (and equivalently the on-chain Aave USDC supply rate, which tracks it), the premium is concentrated in 2024 and falls sharply thereafter (Table 5, Fig. 5): the 2026-to-date carry premium over the risk-free rate is statistically indistinguishable from zero. A linear compression trend, however, is *not* statistically significant over the three-year sample (HAC $t = -1.0$), so we report strong time-variation and a low recent level rather than a confirmed secular decline. A meaningful part of the gross APR is simply the risk-free rate (the $\sim 4.7\%$ T-bill); the genuine risk premium is the $\sim 7\%$ excess, with an excess Sharpe of ~ 5.4 .

5 WHAT DOES NOT BEAT THE BASELINE

We then test five families of overlay. Table 6 summarizes the verdicts.

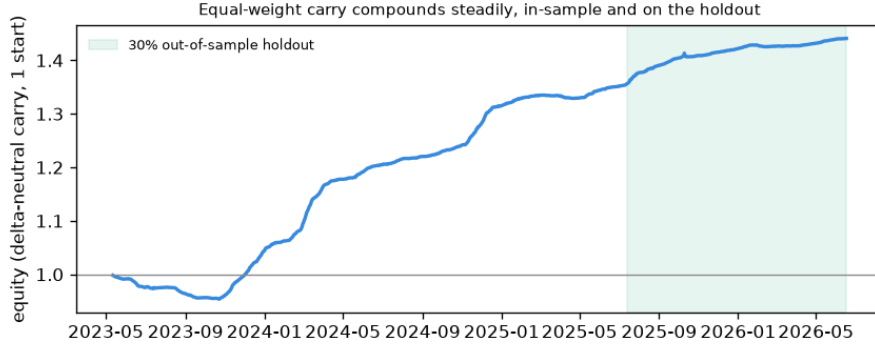


Figure 4: **The payoff.** Equity of the equal-weight delta-neutral carry (\$1 start, fees charged). It compounds steadily and keeps rising on the 30% out-of-sample holdout (shaded), though the slope flattens—visibly consistent with the declining recent premium documented in Table 5.

Table 5: Carry risk premium (excess over T-bill) and HL–Binance funding spread by year, annualized, with Newey–West t -statistics. The $\sim 7\%$ average is dominated by 2024.

	2023	2024	2025	2026 YTD
Carry premium over T-bill (%)	1.1 (ns)	17.9	3.6	−0.8 (ns)
Cross-venue spread (%)	5.8 (ns)	14.7	4.6	2.0

Exit heuristics. A cost-recovery break-even rule with a stop-loss “improved” a recent holdout but is significantly *worse* over the full sample (paired HAC $t = -3.57$, $p < 0.001$; bootstrap $P(\Delta \leq 0) = 1.00$, Sharpe $11.63 \rightarrow 11.18$), winning only one of five walk-forward folds—a textbook holdout-selection artifact.

Cross-sectional allocation. No weighting scheme robustly beats equal weight. Risk-parity (Mailard et al., 2010) appeared significant on the out-of-sample half ($p = 0.03$) but is period-dependent (three of five folds: it helps in the recent compression and hurts in the early bull when high-volatility alts paid). An ML return-tilt is significantly worse.

Tail-hedging. Buying weekly out-of-the-money calls—priced at the *real* DVOL implied volatility—to prevent the short’s liquidation does not improve risk-adjusted return. At prudent leverage liquidations are too rare to protect (premium $\approx 0.16\%$ / year, a wash); at higher leverage the near-ATM weekly premium explodes ($19\%–83\%$ / year), dwarfing the 50bp liquidation penalty. The variance risk premium—implied volatility systematically above realized (Bollerslev et al., 2009), documented for Bitcoin options specifically (Alexander & Imeraj, 2020)—makes buying protection a net cost. Low leverage manages the tail better, and for free.

6 VOLATILITY-TARGETED LEVERAGE: A SMALL REAL REFINEMENT

Volatility is forecastable (EWMA $R^2 \approx 0.33$ for next-day realized vol), and scaling exposure to forecast volatility is known to improve risk-adjusted returns in equities (Moreira & Muir, 2017). The rule is symmetric and scales leverage *inversely* to forecast volatility: it levers *up* when forecast volatility is low and, just as importantly, levers *down* when forecast volatility is high. The de-levering is what does the work—by shedding exposure ahead of turbulent regimes it sidesteps the liquidations a fixed high leverage would suffer, so the extra leverage carried in calm periods is not paid back in blow-ups. Net, one can run about one extra turn of *average* leverage at the same liquidation rate (e.g. $4\times$ vs $3\times$ at ~ 2.5 liquidations/year), lifting return-on-capital from 4.25% to about 4.42%. The gain is bounded by the thin carry yield and by the fact that the liquidating *up-jump* is only partly predictable: 18% of the worst up-moves strike out of below-median volatility, so down-scaling cannot pre-empt every blow-up. Crucially, the refinement is free only if implemented

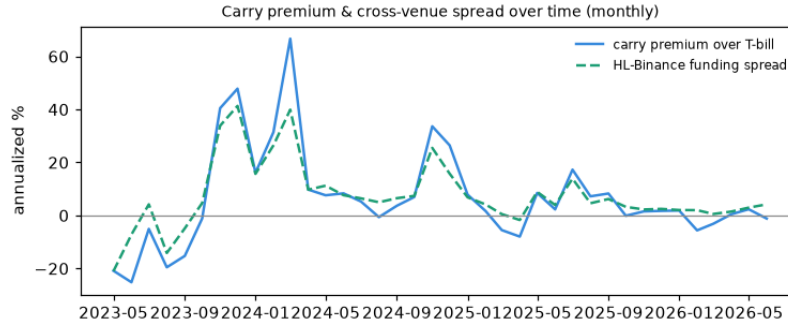


Figure 5: Monthly annualized carry risk premium (over T-bill) and HL–Binance funding spread. Both are concentrated in 2024 and fall to low single digits; the premium oscillates around zero in 2026. The headline $\sim 7\%$ figures are time-varying averages.

Table 6: Overlays tested against the naive carry baseline, with the significance verdict. “Null” denotes no significant improvement; “negative” denotes significantly worse.

Overlay	Idea	Verdict
Cost-recovery heuristic	exit break-even + stop-loss timing	negative (HAC $t = -3.57$, $p < 0.001$)
Cross-sectional allocation	risk-parity / inverse-vol weighting	null (period-dependent, 3/5 folds)
ML return-tilt	weight toward predicted carry	negative (overfits, churns)
ML / tail-risk leverage	size leverage to a liquidation quantile	null at prudent leverage
Options tail-hedge	buy OTM calls to cap liquidations	negative (premium $\gg$ penalty)

by adjusting the *margin buffer* (a collateral transfer); resizing the position to target volatility incurs turnover whose fees ($\sim 0.5\text{pp}$) cancel the gain.

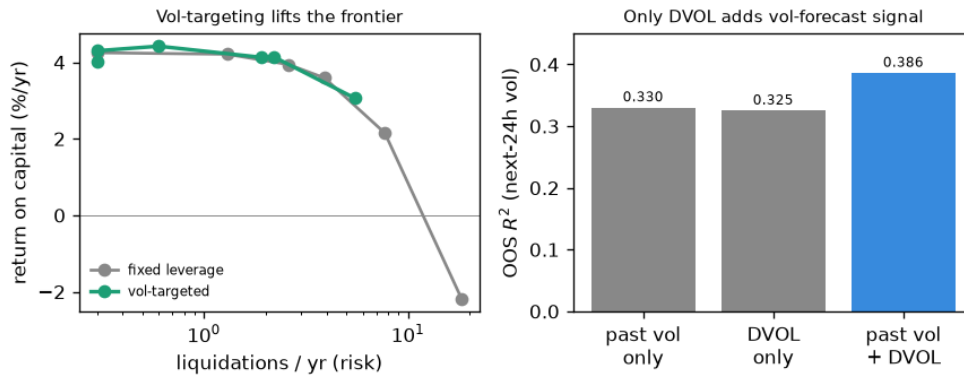


Figure 6: **Volatility tools.** Left: matched-risk efficient frontier (return-on-capital vs liquidations/year)—vol-targeted leverage sits above and to the left of fixed leverage, earning more funding at the same risk (peak RoC 4.42% vs 4.25%). Right: out-of-sample R^2 for predicting next-24h realized volatility; the options-implied DVOL index is the only signal that adds materially over past volatility ($0.330 \rightarrow 0.386$), but it informs the vol *level*, not the up-jump tail that liquidates, so it yields no leverage gain at prudent sizing.

Can a better volatility forecast help? Of all candidate predictors, only the Deribit DVOL implied-volatility index adds real power, improving next-day realized-vol R^2 from 0.330 to 0.386 (+0.056), far above the +0.006 added by funding, basis, and cross-asset-stress features (asymmetric and jump features add essentially nothing)—consistent with the variance-risk-premium evidence that implied volatility carries information beyond realized (Bollerslev et al., 2009; Alexander & Imeraj, 2020). (Our forecasts use gradient-boosted quantile regression, as in recent crypto-volatility work (Dudek et al., 2025; Djanga et al., 2023); the cross-asset volatility feature exploits the intra-day volatility commonality documented by the latter.) Yet this does not become actionable: DVOL improves the *level* of volatility, not the *up-jump tail* that triggers liquidation, and at prudent leverage the baseline already incurs zero liquidations, so feeding DVOL into the leverage rule yields no return-on-capital gain.

7 DISCUSSION

Each overlay we tested embodied a plausible, pre-registered premise. The cost-recovery exit assumed that banking the carry once its costs are recovered, with a stop-loss, beats holding through drawdowns. Cross-sectional allocation assumed that weighting assets by inverse volatility or risk parity beats equal weight. The machine-learning return-tilt assumed funding is predictable enough to overweight the assets about to pay the most. The volatility-targeting overlay assumed volatility is forecastable, so that *cutting* leverage when forecast volatility is high and *raising* it when low lets one run a higher *average* leverage at the same liquidation rate. The tail-risk leverage rule assumed the liquidating up-jump could be anticipated from a quantile model, and the options overlay assumed out-of-the-money calls could cheaply cap that tail. Each premise is individually sensible; the question is which, if any, survives out-of-sample, multiple-testing-corrected scrutiny.

Tested that way, a consistent pattern emerges: *risk-based* sophistication occasionally helps a little, *return-prediction* sophistication reliably overfits and hurts, and the only robust standalone edge is a structural cross-venue dislocation rather than anything intrinsic to the single-venue carry. The naive baseline—buy-and-hold, low fixed leverage, equal weight—is remarkably hard to beat.

Methodologically, the study is a case study in how easily a backtest deceives. The exit heuristic and the risk-parity allocation both produced confident “improvements” on a recent holdout that vanished or reversed under walk-forward and bootstrap testing. The productive findings here—the cross-venue edge, the volatility-targeting bound, and the negative results—came from hypothesis-driven analysis with significance testing and walk-forward built in from the start. In a domain where most plausible ideas overfit, that statistical discipline—pre-registration, HAC inference, walk-forward, and multiple-testing correction—is what separates a real edge from a holdout artifact.

8 CONCLUSION

The Hyperliquid funding carry is a real but highly time-varying risk premium that a simple baseline captures near-optimally; clever overlays mostly do not help and several significantly hurt. The premium averages $\sim 7\%$ over the risk-free rate but is concentrated in 2024 and near zero in 2026 to date (a secular compression trend is not statistically confirmable on three years). The genuine, robust, and economically sensible edge is cross-venue: Hyperliquid funds structurally above Binance, a delta-neutral spread that is highly significant, walk-forward consistent, survives basis, capital, and cost frictions when held statically, and is more persistent than the single-venue premium (still significant in 2026)—though it, too, has declined from its 2024 level. We hope the significance-first methodology, and the catalog of honest nulls, is useful to practitioners tempted by the next plausible refinement.

DATA AVAILABILITY

All inputs are obtained from public sources: Hyperliquid and Binance (funding rates, premia, prices), Deribit (the DVOL implied-volatility index), FRED (the 3-month Treasury-bill rate, series DGS3MO), and DefiLlama (Aave-v3 on-chain supply rates and Hyperliquid TVL). The derived datasets, full analysis code, frozen per-result statistics, and scripts that regenerate every figure and table are available at <https://github.com/intrepidcanadian/projectresearch>.

AUTHOR CONTRIBUTIONS AND USE OF AI

This is a single-author work. The author formulated the research questions, designed and pre-registered the hypotheses, directed and supervised all analyses, interpreted the results, drew the conclusions, and is solely responsible for the content.

AI-assisted technologies were used as tools, under the author’s direction and control, in two roles. First, in *implementing the analysis*: following the research questions, hypotheses, and experiment designs specified by the author, a large-language-model assistant helped write and run the back-testing and statistical code on the cached data; the author admitted no result to the paper that had not passed the pre-registered significance criteria under the author’s review. Second, in the *writing process*: a large language model assisted in drafting and editing the manuscript to improve readability. The author conceived the research questions and hypotheses, critically reviewed and verified all AI-assisted code, statistics, and prose, and is solely responsible for the content. No AI system is an author of, or bears responsibility for, this work.

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